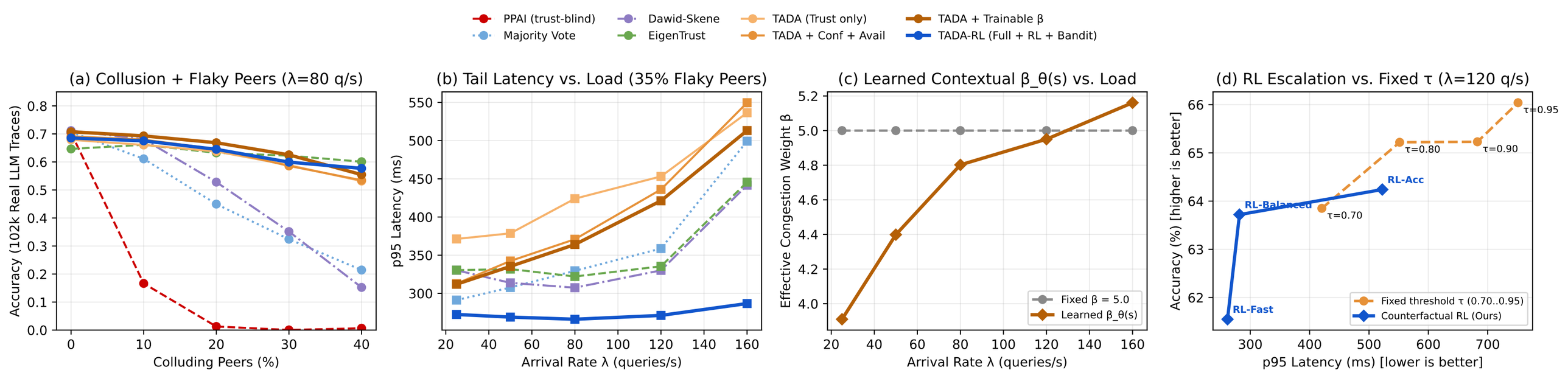
PPAI (trust-blind)
Majority Vote
Dawid-Skene
EigenTrust
TADA (Trust only)
TADA + Conf + Avail
TADA + Trainable β
TADA-RL (Full + RL + Bandit)
(a) Collusion + Flaky Peers (λ=80 q/s)
Accuracy (102k Real LLM Traces)
Colluding Peers (%)
(b) Tail Latency vs. Load (35% Flaky Peers)
p95 Latency (ms)
Arrival Rate λ (queries/s)
(c) Learned Contextual β_θ(s) vs. Load
Effective Congestion Weight β
Arrival Rate λ (queries/s)
Fixed β = 5.0
Learned β_θ(s)
(d) RL Escalation vs. Fixed τ (λ=120 q/s)
Accuracy (%) [higher is better]
p95 Latency (ms) [lower is better]
τ=0.70
τ=0.80
τ=0.90
τ=0.95
RL-Balanced
RL-Acc
RL-Fast
Fixed threshold τ (0.70..0.95)
Counterfactual RL (Ours)